

The Workforce Consequences of Ending Mandatory Union Fees: Evidence from *Janus v. AFSCME**

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Abstract

Public-sector unions shape how state and local governments structure their workforces, but shocks to union bargaining power are rare. The 2018 U.S. Supreme Court ruling in *Janus v. AFSCME* provided one by eliminating mandatory agency fees in 21 states. Using employer-level Census of Governments data and difference-in-differences, I find *Janus* reduced part-time employment by 8.4 percent, raised the full-time share by 1.75 percentage points, and left full-time employment unchanged. Although some specifications suggest a rise in per-worker earnings, a decomposition shows this is a consequence of shedding lower-paid part-time workers rather than wage increases — a result confirmed in CPS data.

Keywords: Public-sector unions; Union membership; Collective bargaining; Agency fees; *Janus*; Free-riding

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1 Introduction

Public-sector unions represent millions of state and local government workers and shape how those workforces are sized, staffed, and compensated. Yet how much of that influence depends on the institutional underpinnings of union bargaining power has been difficult to study, because such shocks are rare — until the 2018 U.S. Supreme Court ruling in *Janus v. AFSCME* delivered an unusually large one. Overturning four decades of precedent, the Court held that public-sector workers could no longer be compelled to pay agency fees as a condition of employment, immediately making union financial support voluntary in the 21 states where it had previously been mandatory. In this paper, I examine how a contraction in union bargaining power reshapes the public-sector workforce — its size, its full-time/part-time composition, and the compensation it receives — for the roughly 6 million state and local employees affected by this landmark ruling (DiSalvo, 2019).

The labor market effects of *Janus* are ambiguous *ex ante*. If union bargaining power falls, wages or employment or both could decline. However, public-sector wage schedules are embedded in multi-year collective bargaining agreements, civil-service pay scales, and political commitments that limit short-run adjustment (Freeman, 1986), insulating wages from immediate change. Adjustments on the employment margin — and particularly on part-time employment, achieved through hiring restraint, the non-renewal of seasonal positions, and attrition — are more politically feasible when union capacity to resist them weakens (Poterba and Rueben, 1995; Scorsone and Plerhoples, 2010). This asymmetry motivates focusing on employment composition as a key margin of adjustment.

That wage-stickiness intuition stands against a standard neoclassical alternative (Farber, 1986), which predicts weaker unions yield lower per-worker compensation and higher employment. It can also be refined further: insider-outsider models (Lindbeck and Snower, 2001; Saint-Paul, 1996; Bentolila and Dolado, 1994) imply that with union political capacity contracted, residual capacity protects full-time ‘insiders’ over contingent ‘outsiders’ — part-time and seasonal workers less represented in union membership and leadership. This refinement also predicts effects to concentrate in governments where union political capacity historically operated through electoral channels (Anzia, 2011, 2013; Sieg and Wang, 2013) and be muted in appointed-board governments. I adjudicate among these predictions using the data in Section 5.

Using data from three waves of the quinquennial Census of Governments (CoG) from 2012, 2017, and 2022 and a differences-in-differences (DiD) approach comparing RTW and non-RTW states I document two key results. First, I find that *Janus* reduced part-time employment by about

8.4 percent in governments in treated states relative to governments in RTW states (control), with no statistically or economically meaningful change in full-time employment. The result of these changes is a rise in the full-time employment share by about 1.75 percentage points. Second, I find no robust impact of the decision on per-worker earnings. Although some specifications do indicate a small *increase* following *Janus*, an earnings decomposition exercise shows that this increase is driven entirely by the mechanical elevation of the per-worker average caused by the exit of lower-paid part-time workers, and *not* by wage increases for continuing employees.

I confirm the decline in part-time employment and the increase in the full-time share using three additional approaches. First, I splice the CoG with the Annual Survey of Public Employment and Payroll (ASPEP) to build a balanced 2011–2025 panel. The annual frequency of this panel also supports an event-study specification, a central test of the parallel trends assumption underlying any DiD design. Second, to isolate the effect of *Janus* from broader local shocks, I compare pairs of bordering states in which one had adopted RTW laws (e.g., Indiana) and its neighbor had not (e.g., Ohio). Third, using the Census Finance Survey, I construct matched difference-in-differences estimators that condition on pre-treatment fiscal stress and liquidity. The employment results hold across all three.

The same three approaches also confirm the earnings null. I further validate that result using 13 waves of the Current Population Survey (CPS) from 2011–2024 (excluding 2018), whose individual-level data on earnings and worker demographics let me corroborate the result in a setting where composition concerns are attenuated. Using the CPS, I also show that employer health insurance and pension provision did not change after *Janus*.

Examining heterogeneity by government type offers clues to the underlying mechanism. The decline is largest in counties (about 12.8 percent in the CoG panel), present in municipalities and school districts, and statistically indistinguishable from zero in special districts, which are run predominantly by appointed boards. This pattern points to a political economy channel: by eroding unions’ electoral capacity, *Janus* lowered the political cost to elected officials of workforce restructuring.

The paper contributes to the literature on public-sector labor market institutions in three ways. First, it identifies an employment-composition margin of adjustment to a major shock to union bargaining power, distinct from the wage and membership margins studied elsewhere. The closest related work is Bagchi (2026), which uses administrative payroll data from 400 state and local governments to study *Janus*’s impact on union membership and earnings growth. The present paper differs in three respects: It covers the universe of state and local governments via the CoG;

it focuses on overall headcount and workforce composition; and it examines earnings systematically using event studies and a Lemieux (2006) decomposition. Second, it shows that public-sector union power affects staffing choices even when wages are sticky, extending Frandsen (2021) and Freeman (1986) to the full-time/part-time margin. Third, it provides evidence consistent with a political economy channel through which agency fees shaped employer behavior: the employment effects concentrate in electorally accountable counties and municipalities and are absent in appointed-board special districts, consistent with unions’ political capacity mattering where elections discipline employers (Flavin and Hartney, 2015; Feigenbaum et al., 2018; Carreri et al., 2025).

The paper proceeds as follows. Section 2 provides institutional background on *Janus* and public-sector collective bargaining. Section 3 describes the data and sample construction. Section 4 outlines the empirical strategy. Section 5 reports the results, including event studies, employment and earnings regressions, and heterogeneity by government type. Section 6 concludes.

2 Institutional Landscape: The *Janus* Decision and Public Sector Collective Bargaining

While the National Labor Relations Act of 1935 governs private-sector bargaining, the rights of public-sector employees in state and local government are determined by state law. Prior to *Janus*, public-sector union law was governed by *Abood v. Detroit Board of Education* (431 U.S. 209, 1977), which held that states could constitutionally require non-union public employees to pay agency fees — also called “fair share” fees — to cover the union’s costs of collective bargaining and contract administration. As of 2018 at the time of the ruling, 21 states that had not enacted RTW laws permitted such fees in public-sector agreements, with agency fees typically set to about 85 percent of full union dues (Murphy, 2019).¹

The 2018 *Janus* ruling, by a narrow 5–4 majority, held that mandatory contributions to a union constituted compelled speech and were therefore unconstitutional under the First Amendment. The decision took immediate effect, rendering mandatory agency fee arrangements in public-sector contracts unenforceable. However, because the ruling left state labor law unaffected, unions in

¹These states are: Alaska, California, Connecticut, Delaware, Hawaii, Illinois, Maine, Maryland, Massachusetts, Minnesota, Montana, New Hampshire, New Jersey, New Mexico, New York, Ohio, Oregon, Pennsylvania, Rhode Island, Vermont, and Washington. Because most other states already prohibited mandatory union payments through RTW laws, *Janus* had no direct impact there. I exclude Colorado as its Labor Peace Act required a 75 percent “super-majority” to authorize agency fees; consequently, most local and all state agencies in Colorado operated as open shops prior to 2018.

affected states retained exclusive representation rights. Thus, unions remained legally obligated to represent all workers in the bargaining unit, including nonmembers, even as they lost the agency-fee revenue that had supplemented member dues, while also facing the prospect that full members might exit as well.

3 Data and Sample Construction

Primary dataset: CoG three-wave panel. My primary dataset uses the 2012, 2017, and 2022 Census of Governments (CoG), which constitute a complete enumeration of approximately 90,000 state and local government units. The 2012 and 2017 waves provide two clean pre-*Janus* observations; the 2022 wave provides a clean post-*Janus* observation four years after the ruling, after the initial adjustment period has passed. These time points enable me to examine whether there were any trends in employment or earnings leading up to *Janus* as well as capturing the medium-run post-treatment period. Because the CoG is a census rather than a sample, the sample preserves the full size distribution of U.S. local governments — including the many small units that make up the majority of the universe. The limitation of the CoG dataset is its three total observations per unit, which constrains dynamic pre-trend testing.

Secondary dataset: Balanced annual CoG-ASPEP panel. To conduct event studies, I supplement the CoG with the Annual Survey of Public Employment and Payroll (ASPEP), a Census Bureau survey of state and local government employment and payroll that, unlike the CoG, provides yearly observations but only covers a subset of the roughly 90,000 government units. I construct a panel by splicing the three CoG years (2012, 2017, and 2022) with the ASPEP survey years. After limiting myself to only including government units that are present in all 15 years from 2011 through 2025 and applying the sample restrictions described below, the panel contains approximately 3,670 government-unit observations per year for a total of roughly 55,000 observations with eight pre-*Janus* and seven post-*Janus* annual observations per unit.²

The balance requirement, combined with the inherent ASPEP sampling, shrinks the panel to roughly one in ten CoG units (3,670 vs. 34,898),³ with the steepest attrition among independent

²The March payroll reference month for the 2018 survey precedes the June 2018 ruling; accordingly, 2018 is counted here as a pre-*Janus* year, though it serves as the omitted reference year in the event-study specification.

³Starting from the approximately 90,000 CoG units, I drop three states with ambiguous agency-fee status (Colorado, Missouri, Wisconsin) and all townships (roughly 14,800 units). The most consequential further restriction — units with positive full-time and part-time employment in all three waves, required by the log specification — excludes 13,783 governments, predominantly small special districts. The resulting estimation sample is 34,898 units as of 2017, of which 47 are state governments included in the overall regression but excluded from the government-type sub-panels (Table 5.4).

school districts, where fewer than 7 percent are retained. Because ASPEP selection tilts toward larger and more full-time-intensive units, the balanced panel differs systematically from the CoG three-wave survey along several dimensions: median total employment is more than ten times higher (602.5 vs. 60), the median FT share is higher at 0.82 versus 0.69, and median full-time earnings are roughly 15 percent higher. I therefore treat the CoG three-wave panel as primary and the balanced panel as secondary, valuing the latter chiefly because it enables dynamic identification.

Sample restrictions. I exclude states with ambiguous agency-fee status before *Janus* (Colorado, Missouri, and Wisconsin), leaving 47 states. Treated states are those with mandatory agency fee statutes in effect prior to the ruling; states that already prohibited mandatory fees serve as the control group. I also drop townships from both datasets – the CoG three-wave survey and the CoG–ASPEP balanced panel, as that category conflates structurally distinct entities across states.⁴

Outcomes. The primary outcomes are the natural logarithm of full-time employment, the natural logarithm of part-time employment, and the full-time employment share, defined as full-time employment divided by total employment. I also examine log total employment and log annual earnings per worker for full-time, part-time, and total employees to analyze if the *Janus* ruling impacted either the employment margin or the earnings margin.

4 Empirical Strategy

As motivated in the introduction, the institutional asymmetry between wage and employment adjustment in the public sector implies that employment composition is the natural margin on which to detect a labor-market response to *Janus*. The baseline estimating equation is:

$$y_{it} = \alpha_i + \gamma_t + \beta(\text{Treated}_s \times \text{Post}_t) + \delta_s \cdot t_{\text{pre}} + \varepsilon_{it}$$

where y_{it} is an outcome for government unit i in year t ; α_i are unit fixed effects, γ_t are year fixed effects, Treated_s equals one for units in states with mandatory agency fees before *Janus*, Post_t equals one for years after 2018, and $\delta_s \cdot t_{\text{pre}}$ is a state-specific linear time trend identified off the pre-*Janus* period only. The coefficient of interest is β . Because the Supreme Court’s ruling took effect across all affected states and, as documented in prior work, payroll departments implemented the decision within a single pay cycle (Bagchi, 2026, Table A.2), all treated states are exposed to the

⁴Full-time employment averages 31 workers per township in affected non-RTW states but fewer than 2 in RTW states — a roughly sixteen-fold difference — indicating that the label captures very different tiers of local government depending on regional history and legal tradition.

shock simultaneously. Consequently, the heterogeneous timing concerns prominent in the recent DiD literature (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020) do not apply.

I augment the standard two-way fixed effects (TWFE) estimator with state-specific pre-period linear trends and use that as my baseline specification to allow for the possibility that treated and control states were on different pre-treatment growth trajectories. This is a relevant concern because agency-fee states tend to have higher unionization, higher per-employee spending, and potentially different long-run trends in public-sector employment and compensation. I cluster standard errors by state throughout.

For the balanced annual panel that I construct by piecing together the CoG with the ASPEP, in addition to estimating the regression specification above, I plot the results from an event-study specification:

$$y_{it} = \alpha_i + \gamma_t + \sum_{k \neq 2018} \beta_k (\text{Treated}_s \times \mathbf{1}[t = k]) + \varepsilon_{it}$$

with 2018 as the omitted reference year. The pre-period coefficients ($\beta_k, k < 2018$) serve as a direct test of the parallel trends assumption: If treated and control units were on similar trajectories before the ruling, these coefficients should be statistically indistinguishable from zero.

All estimates are unweighted. This reflects the estimand of interest: what happened to the average affected local government. Estimates weighted by total employment answer the different question of what happened to the average affected public employee; those weighted estimates are considerably attenuated and corresponding results are included in the Appendix as a distinct estimand.

To cross-validate the earnings null at the individual level, I also estimate DiD models using the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC), restricting to public-sector workers and using individual-level data on annual earnings and hourly wages. I also examine employer-provided health insurance coverage and pension offering as total compensation could adjust on the benefits margin even if wages are rigid. I include individual-level controls (age, education, sex, and full-time status) in the CPS regressions as well as state and year fixed effects, along with state-specific pre-period trends.

5 Results

5.1 Overall changes to employment, earnings, and the full-time share following *Janus*

Table 1 reports estimates for all outcomes involving employment and earnings — separately for full-time, part-time, and overall workers — in addition to the full-time share of employment, using a specification that includes two-way fixed effects and state-specific pre-period trends. Panel A presents results from the CoG three-wave panel (the primary dataset) and Panel B from the balanced annual CoG–ASPEP panel.

The results are consistent across both datasets. I observe that full-time employment is unchanged with the coefficients small and statistically insignificant throughout. In contrast, the data indicate a reduction in part-time employment by about 8.4 percent in the CoG three-wave panel and by about 6.3 percent in the annual panel following *Janus*.⁵ Driven by this decline in part-time employment, total employment falls – by about 2.6 percent in the three-wave panel and about 1.2 percent in the annual panel, although the latter estimate is statistically insignificant. The consequence of a decline in part-time employment without a corresponding change in full-time employment is a rise in the share of employees working full time by about 1.75 percentage points in the three-wave CoG panel — a 2.7 percent increase relative to the pre-period mean full-time share of 0.64 — and by about 1.05 percentage points in the annual balanced panel. The somewhat smaller magnitudes in the annual panel likely reflect averaging over years in which the initial adjustment was still underway, whereas the three-wave CoG estimate is based on 2022 when much of the adjustments had already taken place.⁶

Turning to earnings, I find no robust change in earnings per worker for either full-time or part-time workers. Total earnings appear to hold steady in the balanced annual panel, while the CoG three-wave panel suggests a small and statistically significant increase of about 1.8 percent. As I show subsequently, this increase reflects the mechanical elevation of the per-worker average caused by the exit of lower-paid part-time workers and not by wage growth for continuing employees.

Interacting the treatment with an above-median public-sector union density indicator (2017

⁵The decline in PT employment operates on the intensive margin — treated governments reduced PT headcount rather than eliminating PT positions entirely — so the log specification, which drops zero-employment observations, does not bias the estimated effect, as confirmed by the IHS robustness check in Table A.3.

⁶Occupational heterogeneity analysis using CoG and CoG-ASPEP function-level data reveals broadly uniform effects across occupational categories, with the notable exception of fire protection, where pre-existing differential staffing trends between treated and control states preclude a clean *Janus* estimate.

CPS), I find the employment-level declines concentrated in low-density treated states — consistent with stronger unions retaining greater capacity to resist staffing adjustments — while the full-time share increase is uniform across the two groups.

5.2 Identification

A possible concern with the estimates in Table 1 is that the post-2018 declines reflect differential pre-*Janus* trends rather than the ruling itself. Figure 1 presents event-study estimates for log part-time employment, log full-time employment, and the full-time employment share using the balanced annual panel. Because the event study is estimated on the CoG–ASPEP panel rather than the CoG three-wave sample, it serves as a test of parallel trends and a characterization of adjustment timing, not a magnitude comparison with the headline DiD estimates.

For part-time employment, the pre-*Janus* coefficients (2011–2017) are small, close to zero, and statistically indistinguishable from zero, providing no evidence of differential pre-treatment trends. Post-*Janus*, the estimates turn modestly negative in 2019 and 2020 before deepening sharply in 2021 to approximately -10 log points ($p < 0.01$), then partially recovering to around -5 to -8 log points through 2024; by 2025 the estimate attenuates further and is no longer statistically significant. This pattern is consistent with an initial adjustment concentrated in 2021 — as governments worked through budget cycles and attrition — followed by partial mean-reversion as hiring resumed. For full-time employment, coefficients are small and statistically insignificant throughout the entire sample period — pre and post alike — with no systematic directional pattern, confirming that *Janus* did not generate a meaningful change in full-time employment levels. The full-time share tells the complementary story: pre-period coefficients from 2011 through 2017 are flat and near zero, then shift persistently positive from 2021 onward, settling in the range of 1.0 to 1.2 percentage points above the reference year through 2024 before attenuating in 2025. This sustained post-period rise is consistent with the selective loss of part-time workers, with no corresponding change on the full-time margin. I include results from the underlying regressions in Table A.6.

In the CoG three-wave panel, with only two pre-period observations, the state-specific linear trend is exactly identified. As a placebo, I replace the binary post indicator with separate 2012 and 2022 dummies, each measured relative to 2017. For part-time employment, the 2012 placebo is small and statistically indistinguishable from zero while the 2022 coefficient remains negative and significant. For the full-time share, the residual post-*Janus* effect is substantial even after

subtracting a linear extrapolation of the pre-period trend. The wave-specific estimates locate the compositional shift in the post-*Janus* period rather than reflecting a trend already underway

5.3 Robustness checks: Confirming null on FT employment, decrease in PT employment, and increase in the FT share

Table 2 presents a set of robustness checks that confirm the null effect of *Janus* on full-time employment, a decrease in the part-time employment, and an increase in the share of employees working full-time.

Border discontinuity. In this robustness check, I restrict the sample to government units located in states that share a border with at least one non-agency-fee state, absorbing state-border-pair fixed effects to compare treated-state governments directly to their neighbors across the state line. The results are consistent with the full-panel findings: part-time employment falls by about 6.78 percent, the full-time share rises, and full-time employment remains statistically indistinguishable from zero. Because the border sample contains only 23 state clusters that may be too few for reliable asymptotic cluster inference (Cameron, Gelbach, and Miller, 2008), I confirm these results using wild cluster bootstrap p-values and find that the estimates for both part-time employment and the full-time share remain significant. The border estimates therefore rule out broad regional shocks as a confounding explanation.

Matched difference-in-differences. I match treated and control units on two pre-treatment fiscal variables using nearest-neighbor Mahalanobis matching with exact matching on government type: long-term debt per employee, proxying long-run solvency (Gorina, Maher, and Joffe, 2018), and cash and deposits per employee, proxying cash solvency (Scorsone and Plerhoples, 2010). Both capture dimensions of fiscal stress along which governments face stronger pressure to reduce labor costs once union capacity to resist weakens. Because matching absorbs pre-existing differences in fiscal conditions but does not guarantee parallel trends in employment outcomes, I estimate the matched DiD with the same TWFE and state-specific pre-period trends as used in the baseline.

The results are consistent across all three covariate matching schemes. Matching on long-run solvency alone, part-time employment falls by about 6.6 percent and the full-time share rises by 1.40 percentage points. Matching on cash solvency alone yields a decline in part-time employment of about 7.4 percent and an increase in the full-time share of about 1.45 percentage points. Finally, matching on both dimensions jointly produces the largest estimates: a part-time decline of about 8.7 percent and an increase in the full-time share of 1.61 percentage points. All three matched

designs continue to point to a null effect on full-time employment consistent with the baseline results.⁷

COVID-19 and post-period timing. The post-*Janus* period (2022 in the CoG three-wave panel; 2019–2025 in the annual panel) overlaps with the COVID-19 pandemic, raising the concern that the compositional shift reflects pandemic-era workforce disruption rather than a response to the ruling. Three features of the evidence argue against this interpretation. First, Figure A.3 shows that the full-time employment share begins rising in treated counties and municipalities as early as 2019 — a full year before the pandemic. The 2020 observation also predates meaningful disruption, given that the ASPEP measures employment for the pay period containing March 12 and thus captures the workforce prior to the declaration of the national emergency. Second, full-time employment is statistically indistinguishable from zero throughout the post-period (Figure 1, Panel A), inconsistent with a narrative solely based on broad pandemic-era contractions. Third, and perhaps most tellingly, special districts operate in the same regional labor markets as counties and municipalities yet show no post-*Janus* break in either the full-time share (Figure A.3, Special Districts panel) or part-time employment. This within-market contrast rules out COVID-19 as the primary explanation.

Robustness to functional form. Replacing the log transformation with the inverse hyperbolic sine — which retains zero-employment observations — yields estimates that are virtually identical to the baseline, confirming that the results are not driven by the exclusion of zeros from the log specification.

Robustness to use of alternate datasets. As a placebo check, I estimate the identical specification on private-sector employment using QCEW state-level annual data. Figure A.1 shows that private-sector part-time employment and the full-time employment share display neither pre-trends nor a post-*Janus* break, consistent with the public-sector results reflecting a *Janus*-specific compositional shift rather than broad regional labor market trends.

5.4 Government-Type Heterogeneity and the Political Economy Channel

The within-market contrast highlighted above points to governance structure as the key source of heterogeneity. Tables 3 and A.7 report estimates of the baseline specification estimated separately for each government type — the CoG three-wave and annual panels, respectively — and the results reveal a pattern consistent with a political economy channel operating through electoral

⁷Results from border discontinuity and matched DiD regressions using the CoG–ASPEP balanced panel are included in Table A.5.

accountability.

Among the various forms of local government, counties show the largest and most precisely estimated part-time decline — about 12.8 percent in the CoG three-wave panel and about 8.5 percent in the balanced annual panel, with a null effect on full-time employment in both datasets. The share of full-time employees also rises by about 1.3 percentage points in the CoG panel and about 0.9 percentage points in the annual panel. Municipalities show the same pattern: part-time employment falls by about 6.6 percent in the CoG panel and 8.9 percent in the annual panel, and the full-time share rises by about 1.3 and 1.1 percentage points respectively.

Independent school districts present a somewhat more mixed picture. In the CoG three-wave panel, part-time employment declines by about 6.9 percent and the full-time share rises by a statistically significant 1.7 percentage points. The corresponding estimates in the balanced annual panel are weaker, with a part-time decline that is statistically indistinguishable from zero and a full-time share increase of 1.4 percentage points that falls short of conventional significance levels. However, the balanced panel retains fewer than 5 percent of treated CoG districts and is therefore, unlikely to be representative of the broader universe of school districts. I treat the CoG three-wave estimates as the more reliable evidence for this government type.

The results for special districts stand in sharp contrast. These entities — water and sewer authorities, housing authorities, transit agencies — are governed predominantly by appointed rather than elected officials, and across all specifications and both datasets, the part-time employment and full-time share coefficients are statistically indistinguishable from zero. One might worry that this null reflects differences in baseline workforce composition rather than governance structure. Table A.1 addresses this concern directly: special districts entered the post-*Janus* period with a mean full-time employment share of 0.59 — nearly identical to municipalities at 0.57 — ruling out composition as an explanation for the divergence in post-*Janus* outcomes.

Prior work establishes that the distinction between elected and appointed governance has systematic consequences for policy outcomes. Besley and Coate (2003) show that elected regulators set rates more favorable to consumers than appointed regulators; Whalley (2013) finds that cities with appointed treasurers pay lower borrowing costs than those with elected treasurers; and Bagchi (2019) finds that political competition raises the generosity of defined-benefit pensions for general-purpose governments but not for municipal authorities headed by appointed boards. The contrast between electorally accountable counties and municipalities on one side and insulated special districts on the other is consistent with this body of evidence and points to similar underlying mechanisms.

Against that backdrop, the null result for special districts is consistent with a theory in which agency fees mattered in part because they funded the political capacity that disciplines electorally accountable employers (Flavin and Hartney, 2015; Feigenbaum et al., 2018; Carreri et al., 2025). The estimated post-*Janus* employment effects are concentrated in counties and municipalities and absent in special districts, although a pooled triple-interaction test of the elected-versus-appointed differential is not precise enough to reject equality of the two effects at conventional levels.

5.5 Earnings Decomposition

The CoG three-wave panel results show a small positive post-*Janus* increase in total per-worker annual earnings of about 1.8 percent. An increase in earnings is surprising; if *Janus* weakened union bargaining power as surmised, one would expect downward pressure on earnings rather than earnings growth. Table 4 resolves this tension by showing that the increase is a mechanical composition effect driven by the loss of part-time employees rather than a genuine wage gain.

I construct two counterfactual earnings series following Lemieux (2006). The composition counterfactual holds the earnings per full-time and per part-time employee fixed at their 2017 (pre-*Janus*) levels and asks how much average per-worker annual earnings would change solely from the post-*Janus* shift in the full-time/part-time employment mix. The wage counterfactual, by contrast, holds the 2017 full-time/part-time employment mix fixed and asks how much average per-worker earnings would change solely from changes in category-specific earnings. Because part-time employees earn substantially less than full-time employees,⁸ a shift toward full-time employment would mechanically raise the per-worker average earnings even if no individual's earnings changed.

The decomposition in Table 4 confirms the composition channel as the operative mechanism. In the CoG three-wave panel, the composition counterfactual is positive and precisely estimated at about 1.7 percent, while the wage counterfactual is economically small and statistically indistinguishable from zero. The balanced annual panel tells the same story: the composition counterfactual is positive and statistically significant at about 1.1 percent, while the wage counterfactual is again small and insignificant. Thus, the positive earnings result that appears in Table 1 reflects the mechanical elevation of the per-worker average caused by the relative contraction of the part-time workforce and not by wage increases for continuing employees.

The decomposition by government type reinforces this interpretation. For counties and munic-

⁸In the CoG three-wave estimation sample, mean annual earnings for part-time employees (\$12,194) are approximately 24 percent that for full-time employees (\$50,703).

ipalities, the composition counterfactual is positive and statistically significant in both datasets — ranging from about 0.7 to 2.1 percent — while the wage counterfactual is uniformly small and insignificant. For special districts, neither counterfactual is statistically significant in either dataset, consistent with the null employment results for that type of government.

School districts offer somewhat mixed results. The composition counterfactual is positive and significant in both datasets (approximately 2.1 percent in the CoG panel and 1.8 percent in the annual panel), consistent with the shift documented for counties and municipalities. However, the wage counterfactual is also positive and marginally significant (+3.4 percent, $p < 0.10$) in the annual panel, but an event study (Figure A.2) shows that this reflects pre-existing convergence: treated districts had relatively lower wages in the early 2010s, and that gradual catching-up continues smoothly through the post-*Janus* period without a visible break at 2019.⁹ The wage estimate is therefore best read as long-run convergence — potentially compounded by the national teacher shortage (Sutcher, Darling-Hammond, and Carver-Thomas, 2016; García and Weiss, 2019) — and not a *Janus* effect. The composition channel, which has flat pre-period trends and is unaffected by this confound, points to the same FT/PT reallocation seen elsewhere.

5.6 Individual-Level Evidence from the CPS-ASEC

As a complementary individual-level test of the composition-channel interpretation, I estimate difference-in-differences models on the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC), restricting the sample to state and local government workers. Because the CPS measures earnings at the worker level, a flat individual-level estimate would corroborate the decomposition’s conclusion that the per-worker increase reflects FT/PT reallocation rather than wage growth for continuing employees.

The wage outcomes I choose are the log of annual earnings and log of hourly wages, each measured either as-is or winsorized at the 5th and 95th percentiles. I present results from a parsimonious specification that includes no controls and then progressively introduce individual-level demographic controls available in the CPS — a strength of this data source over the CoG or the ASPEP. Across all earnings and wage measures and across all specifications, the coefficient on the interaction term of interest, “Treated $\times$ Post,” is small and consistently indistinguishable from zero, as shown in Table A.8.

Another strength of the CPS is that it enables me to examine whether there were changes

⁹State-specific pre-period trends absorb part of this differential, but the underlying trajectory is nonlinear.

in non-wage compensation — such as pensions and health benefits — both of which are roughly double for health insurance and triple for retirement in the public sector than the private sector (Gittleman and Pierce, 2012). The non-wage results reinforce this conclusion: linear probability model (LPM) estimates and probit estimates for whether health insurance coverage is offered and whether pension benefits are offered are indistinguishable from zero across all specifications, as shown in Table A.9.

As a further check using an entirely independent data source, Table A.10 uses BLS QCEW state-level data (2011–2024) to confirm that neither state nor local government employment nor average weekly wages in treated states deviated from the control-state trajectory following *Janus* — under either the baseline specification or the simpler TWFE. Taken together, the evidence confirms that *Janus* did not generate detectable changes in wages or benefit provisions among public-sector workers in affected states and is consistent with the interpretation that the ruling’s primary effect was on the employment-composition margin — specifically, the substitution of full-time for part-time workers — rather than through changes in the terms of employment for continuing workers.

6 Discussion and Conclusions

The *Janus v. AFSCME* ruling was the most significant shock to public-sector labor relations in over four decades. It immediately affected about 6 million workers spread over roughly 32,000 state and local governments. Using high-quality administrative data, I find that the elimination of mandatory agency fees reshaped local government workforces through changes to employment composition rather than through broad headcount cuts or changes to compensation. In the three-wave CoG panel, governments in affected states reduced part-time employment by about 8.4 percent relative to governments in RTW states, while full-time employment was unchanged and the full-time employment share rose by 1.75 percentage points. These results are confirmed in a balanced annual panel with clean event-study pre-trends, and are robust to border discontinuity designs and difference-in-differences estimators that use covariate matching.

In terms of the earnings impact, I do not observe any statistically significant change for either part-time or full-time workers, with the rise in per-worker earnings for the overall sample in the three-wave panel explained entirely by the mechanical compositional effect of losing lower-paid part-time workers. I confirm these results using the CPS that lets me control for individual-level demographic characteristics. The same data source also lets me rule out changes to benefits such

as the provision of health insurance or pension coverage.

The government-type pattern speaks to the possible channels that underpin these results. The part-time decline is driven by counties and municipalities, which are governments headed by elected officials directly accountable to voters, and is absent for special districts governed by appointed boards. This contrast, predicted by the theory of electoral discipline, suggests that agency fees may have mattered in part through the political capacity they provided to unions in dealing with elected officials. The *Janus* ruling, by making mandatory agency fees illegal and allowing employees to opt out of paying union dues altogether, reduced the ability of unions to influence employment outcomes where electoral accountability had previously made union influence operative.

Two qualifications merit emphasis. First, the design identifies the joint effect of agency-fee elimination and the broader institutional response — by unions, by newly empowered employees, and by state governments that adjusted opt-out procedures — and cannot separately attribute the composition shift to any single mechanism within the *Janus* shock. Second, the findings are specific to U.S. state and local governments after 2018 and may not generalize to other public-sector settings. Subject to those qualifications, the evidence establishes that public-sector union power affects more than wages: it shapes the composition of public workforces, particularly the FT/PT margin in electorally accountable governments. The employment composition channel complements the membership response documented in Bagchi (2026); together, they suggest that the consequences of *Janus* extend well beyond dues revenue — reshaping not just union finances but the structure of the public workforces unions have long helped to organize.

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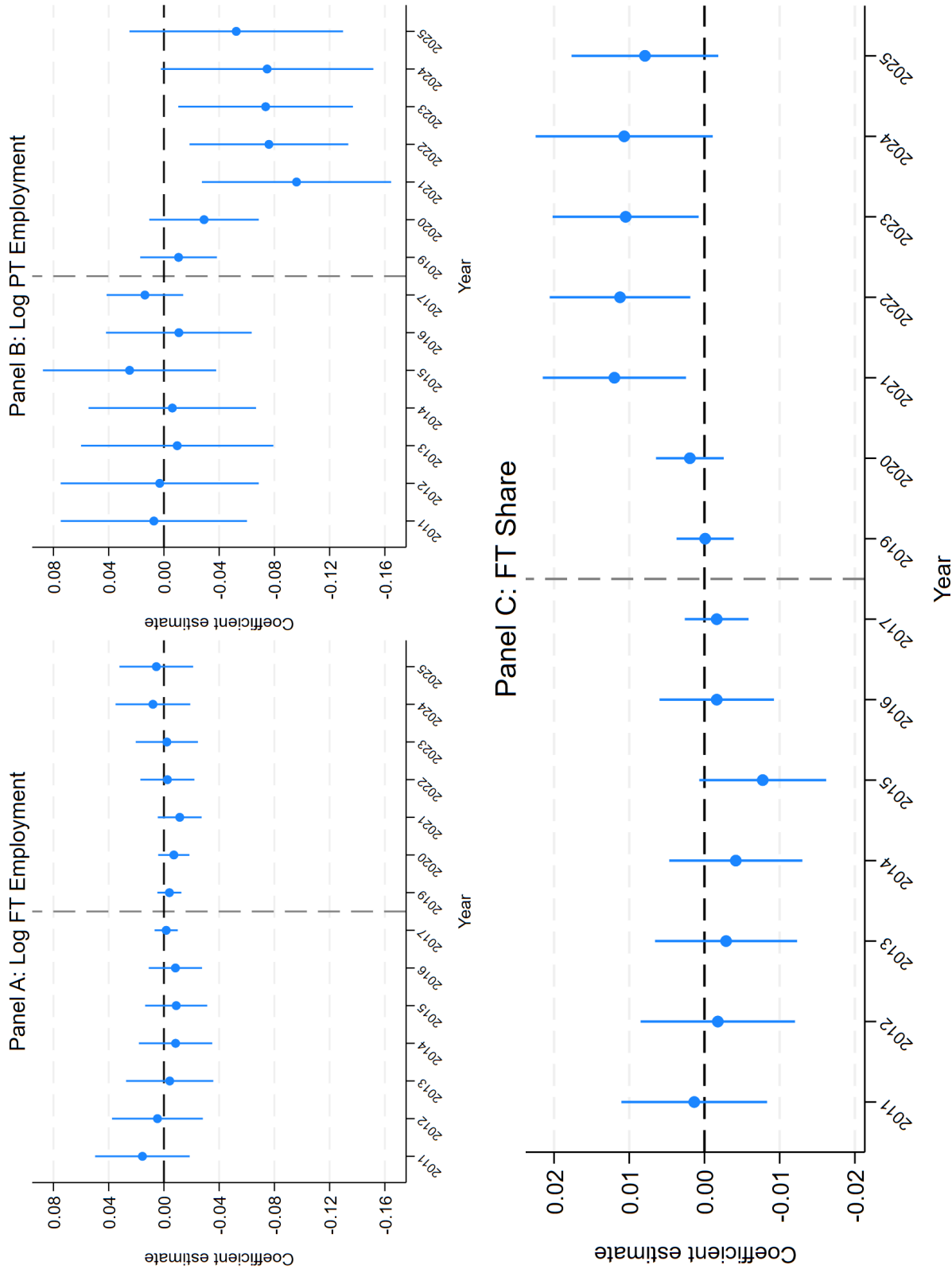
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Figure 1: Event study plot of the effects of *Janus* on FT employment, PT employment, and FT share

Event Study: Employment Composition Effects of *Janus* v. *AFSCME*



2018 is the reference year (pre-*Janus*)

Notes: Each panel plots event-study coefficients from a regression of the outcome on interactions of the treatment indicator with year dummies, relative to the base year 2018. The sample is the CoG-ASPEP annual balanced panel (2011–2025), restricted to units observed in all 15 years; townships and state-government units are excluded. The regressions include government-unit fixed effects and year fixed effects. The treatment indicator equals one for units in states that had a mandatory agency-fee law at the time of the *Janus* ruling. Vertical dashed line marks June 2018 (*Janus* decision). Horizontal dashed line at zero. Regressions are unweighted. The figure displays point estimates and 95 percent confidence intervals; standard errors are clustered by state.

Table 1: Baseline estimates of the impact of *Janus* on employment and earnings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Using the three-wave Census of Governments (CoG) for 2012, 2017, and 2022							
Dependent variable	FT Empl.	PT Empl.	Total Empl.	FT Earnings	PT Earnings	Total Earnings	Full-time share
Coefficient on interaction term, Treated X Post	0.00885 (0.0119)	-0.0838*** (0.0175)	-0.0262*** (0.00932)	-0.00489 (0.00754)	0.00513 (0.0101)	0.0178** (0.00781)	0.0175*** (0.00382)
Number of observations	102241	102241	102241	102241	102241	102241	102241
Number of clusters	47	47	47	47	47	47	47
Panel B: Using the balanced panel constructed from the CoG and ASPEP spanning 2011 – 2025							
Coefficient on interaction term, Treated X Post	0.00443 (0.0117)	-0.0635** (0.0263)	-0.0137 (0.0101)	-0.00442 (0.00762)	0.0209 (0.0181)	0.00806 (0.00834)	0.0113*** (0.00395)
Number of observations	55029	55029	55029	55029	55029	55029	55029
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes:The dependent variables are the natural logarithm of full-time employment (FT Empl.), part-time employment (PT Empl.), and total employment (Total Empl.); the natural logarithm of annualized full-time payroll per worker (FT Earnings), part-time payroll per worker (PT Earnings), and total payroll per worker (Total Earnings); and the share of full-time workers in total employment (FT Share). The treatment indicator equals one if the unit is located in a state that had a mandatory agency-fee law in place at the time of the *Janus* ruling (June 2018) and zero otherwise; Colorado, Missouri, and Wisconsin are excluded because their laws changed during the sample period. The CoG three-wave columns use the Census of Governments three-wave panel (survey years 2012, 2017, and 2022); Treated X Post equal to one in 2022 for treated units. The CoG–ASPEP columns use the annual balanced panel (2011–2025); Treated X Post equal to one in 2019 and later for treated units. Both panels exclude townships. The regression sample in each panel is the set of observations with non-missing values for all seven outcomes, determined by the full-time-share specification. All regressions include government-unit fixed effects, year fixed effects, and state-specific pre-period linear trends. Regressions are unweighted.

Table 2: Difference-in-Differences estimates of the impact of *Janus* on employment and earnings using border-state matching and covariate matching approaches on the CoG three-wave dataset

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Using a border-state matching approach							
Dependent variable	FT Empl.	PT Empl.	Total Empl.	FT Earnings	PT Earnings	Total Earnings	Full-time share
Coefficient on interaction term, Treated X Post	-0.0134 (0.0210)	-0.0678* (0.0300)	-0.0303 (0.0229)	0.0160 (0.0136)	-0.0155 (0.0132)	0.0246** (0.0107)	0.00958* (0.00478)
Wild-cluster p-value	0.587	0.054	0.290	0.299	0.260	0.029	0.062
Number of observations	65385	65385	65385	65385	65385	65385	65385
Number of clusters	23	23	23	23	23	23	23
Panel B: Using log of long-term debt per employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	-0.00190 (0.0118)	-0.0659*** (0.0228)	-0.0289*** (0.00977)	-0.00643 (0.00817)	0.000485 (0.0111)	0.0134* (0.00774)	0.0140*** (0.00422)
Number of observations	57172	57172	57172	57172	57172	57172	57172
Number of clusters	47	47	47	47	47	47	47
Panel C: Using log of cash reserves and deposits per employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	-0.00159 (0.0111)	-0.0740*** (0.0172)	-0.0324*** (0.00831)	0.00186 (0.00774)	0.00722 (0.0113)	0.0218*** (0.00794)	0.0145*** (0.00357)
Number of observations	67555	67555	67555	67555	67555	67555	67555
Number of clusters	47	47	47	47	47	47	47
Panel D: Using log of long-term debt and cash reserves/ employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	0.000454 (0.0102)	-0.0871*** (0.0180)	-0.0287*** (0.00888)	-0.000736 (0.00798)	-0.00606 (0.0105)	0.0183** (0.00721)	0.0161*** (0.00387)
Number of observations	54322	54322	54322	54322	54322	54322	54322
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: All panels use the Census of Governments three-wave panel (survey years 2012, 2017, and 2022), excluding townships. Treated X Post equal to one in 2022 for treated units. The dependent variables are the same seven outcomes as in Table 1. Panel A (state-border-pair DiD) restricts to government units in counties that straddle a treated/control state border and includes unit fixed effects, year fixed effects, and state-border-pair segment fixed effects; the comparison group is border-county units on the control side of each state pair. Significance stars are based on wild cluster bootstrap (WCB) p-values using Webb (2023) weights, 999 replications, seed 12345; asymptotic standard errors clustered by state are shown in parentheses for reference. Panels B–D (matched DiD) use 1-nearest-neighbor Mahalanobis matching exact on government type; matching covariates vary by sub-panel: long-run solvency (Panel B), cash solvency (Panel C), or both (Panel D). Matched DiD estimates are shown for the preferred specification with two-way fixed effects (unit and year fixed effects) and state-specific pre-period linear trends, matching the baseline approach in Table 1. The regression sample in each panel is the set of observations with non-missing values for all seven outcomes, determined by the full-time-share specification. All regressions are unweighted.

Table 3: Difference-in-Differences estimates of the impact of *Janus* on employment and earnings – Split by type of government using the Census of Governments

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable	FT Empl.	PT Empl.	Total Empl.	FT Earnings	PT Earnings	Total Earnings	Full-time share
Panel A: Counties							
Coefficient on interaction term, Treated X Post	-0.00748 (0.0115)	-0.128*** (0.0298)	-0.0226* (0.0128)	-0.00216 (0.00733)	-0.0107 (0.0198)	0.00769 (0.00684)	0.0127*** (0.00376)
Number of observations	8171	8171	8171	8171	8171	8171	8171
Number of clusters	45	45	45	45	45	45	45
Panel B: Municipalities							
Coefficient on interaction term, Treated X Post	-0.00196 (0.0140)	-0.0656*** (0.0285)	-0.0270*** (0.0131)	-0.00240 (0.0107)	0.000130 (0.0176)	0.0139 (0.0111)	0.0132*** (0.00596)
Number of observations	37453	37453	37453	37453	37453	37453	37453
Number of clusters	47	47	47	47	47	47	47
Panel C: School Districts							
Coefficient on interaction term, Treated X Post	0.00364 (0.0120)	-0.0687*** (0.0246)	-0.0268*** (0.0110)	0.00597 (0.0107)	-0.00466 (0.0199)	0.0247** (0.0113)	0.0168*** (0.00413)
Number of observations	33621	33621	33621	33621	33621	33621	33621
Number of clusters	44	44	44	44	44	44	44
Panel D: Special Districts							
Coefficient on interaction term, Treated X Post	0.0136 (0.0191)	-0.0391 (0.0263)	-0.0168 (0.0237)	-0.00700 (0.0111)	0.0485 (0.0321)	0.0195 (0.0164)	0.00884 (0.00559)
Number of observations	22855	22855	22855	22855	22855	22855	22855
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: Each panel presents results from a specification with two-way fixed effects and state-specific linear pre-trends estimates for a single government type, estimated separately within that type, using the Census of Governments three-wave panel (2012, 2017, 2022). Treated X Post equals one in 2022 for treated units. Government types are County (Panel A), Municipality (Panel B), Independent School District (Panel C), and Special District (Panel D). State-government units are excluded from this analysis given their limited power but are included in the overall analysis reported in Table 1. Townships are excluded from all panels. Within each type-dataset cell, the regression sample is the set of observations with non-missing values for all seven outcomes, determined by the full-time-share specification. All regressions include government-unit fixed effects, year fixed effects, and state-specific pre-period linear trends. Regressions are unweighted.

Table 4: Decomposition of earnings increase – compositional counterfactual (holding wages steady) and wage counterfactual (holding composition steady)

	(1)	(2)	(3)	(4)	(5)	(6)
	Actual change in earnings	Composition effect	Wages effect	Actual change in earnings	Composition effect	Wages effect
	Panel A1: Overall Sample – CoG dataset					
Coefficient on interaction term, Treated X Post	0.00654 (0.00803)	0.0173*** (0.00441)	-0.00784 (0.00664)	0.00639 (0.00832)	0.0105** (0.00428)	-0.00354 (0.00773)
Number of observations	151130	151130	151130	58426	58426	58426
Number of clusters	47	47	47	47	47	47
	Panel B1: Counties – CoG dataset					
Coefficient on interaction term, Treated X Post	0.00669 (0.00763)	0.0103** (0.00393)	-0.00208 (0.00682)	0.00422 (0.00945)	0.00725** (0.00335)	-0.00458 (0.00972)
Number of observations	8340	8340	8340	11415	11415	11415
Number of clusters	45	45	45	45	45	45
	Panel C1: Municipalities – CoG dataset					
Coefficient on interaction term, Treated X Post	-0.00364 (0.0148)	0.0212*** (0.00751)	-0.00764 (0.0121)	-0.0119 (0.00916)	0.0113* (0.00662)	-0.0204* (0.0116)
Number of observations	50997	50997	50997	20580	20580	20580
Number of clusters	47	47	47	47	47	47
	Panel D1: School districts – CoG dataset					
Coefficient on interaction term, Treated X Post	0.0256** (0.0114)	0.0205*** (0.00483)	0.00606 (0.0109)	0.0497** (0.0198)	0.0182** (0.00811)	0.0338* (0.0187)
Number of observations	36782	36782	36782	11669	11669	11669
Number of clusters	44	44	44	41	41	41
	Panel E1: Special districts – CoG dataset					
Coefficient on interaction term, Treated X Post	0.0117 (0.0169)	0.00617 (0.00716)	-0.00210 (0.0166)	-0.00282 (0.0123)	0.00359 (0.00783)	-0.00853 (0.0115)
Number of observations	54870	54870	54870	14057	14057	14057
Number of clusters	47	47	47	47	47	47
	Panel E2: Special districts – CoG-ASPEP dataset					
Coefficient on interaction term, Treated X Post	0.0117 (0.0169)	0.00617 (0.00716)	-0.00210 (0.0166)	-0.00282 (0.0123)	0.00359 (0.00783)	-0.00853 (0.0115)
Number of observations	54870	54870	54870	14057	14057	14057
Number of clusters	47	47	47	47	47	47

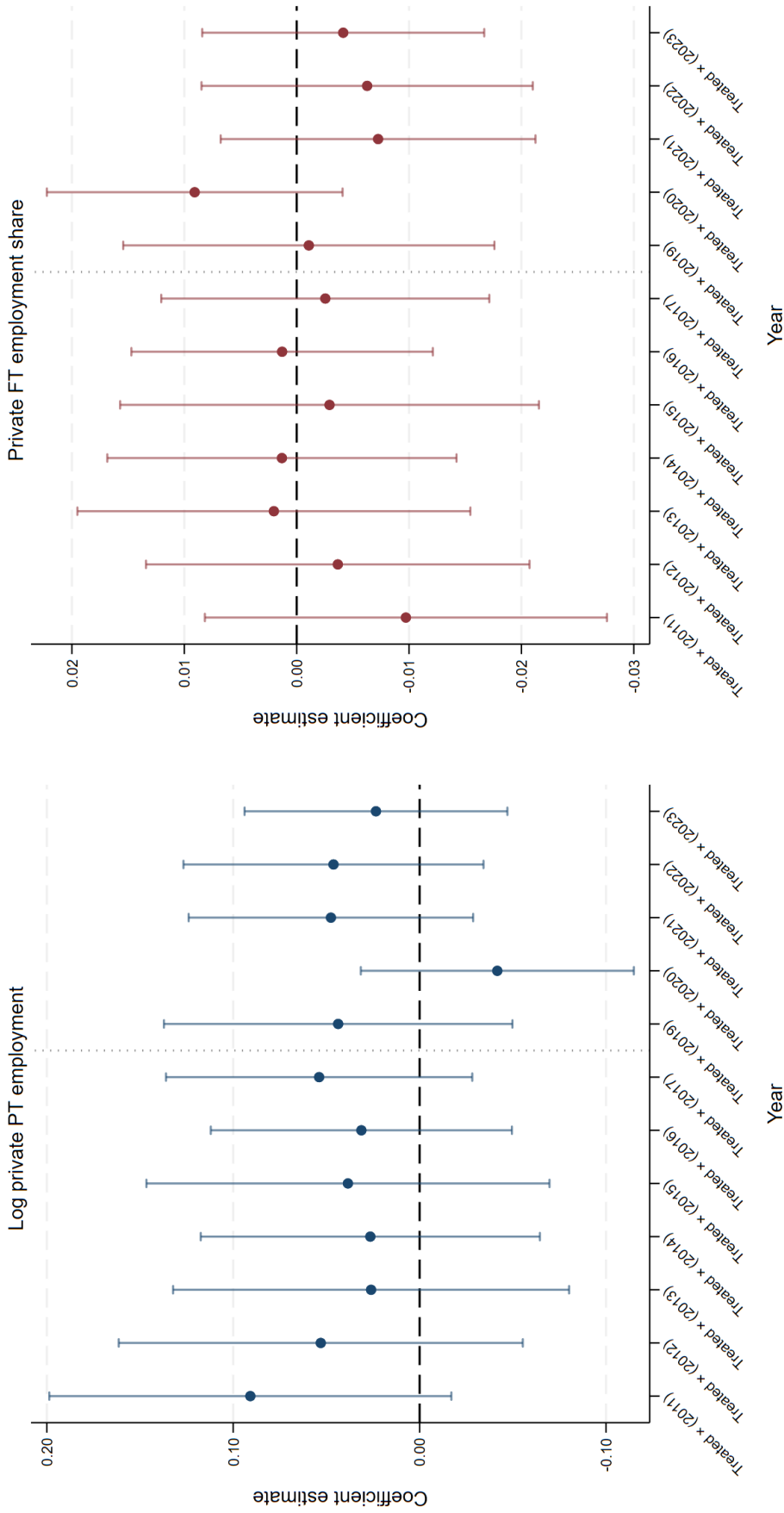
Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: Each panel decomposes the post-*Janus* change in log total per-worker earnings into two counterfactuals following Lemieux (2006). The “Actual” column reports the coefficient on log total annual earnings. The “Composition effect” column is the log of what total earnings would have been had wages remained at their 2017 levels while composition changed; a significant coefficient indicates a composition effect. The “Wages effect” column is the log of what total earnings would have been had the full-time/part-time composition remained at its 2017 level while wages changed; a significant coefficient indicates a wage effect. Columns (1)–(3) use the Census of Governments three-wave panel (2012, 2017, 2022) with Treated X Post equal to one in 2022 for treated units. Columns (4)–(6) use the CoG-ASPEP annual balanced panel (2011–2025) with Treated X Post equal to one in 2019 and later for treated units. Panels A–E present results categorized by government type (County, Municipality, Independent School District, Special District, and All Types, respectively). Within each panel, sub-panels 1 and 2 report estimates using the Census of Governments (CoG) and the Annual Survey of Public Employment and Payroll (CoG-ASPEP) datasets, respectively. Within each panel, the regression sample is the set of observations with non-missing values for all three decomposition outcomes, determined by the total-earnings specification. All regressions include government-unit fixed effects, year fixed effects, and state-specific pre-period linear trends. Regressions are unweighted.

Online Appendix

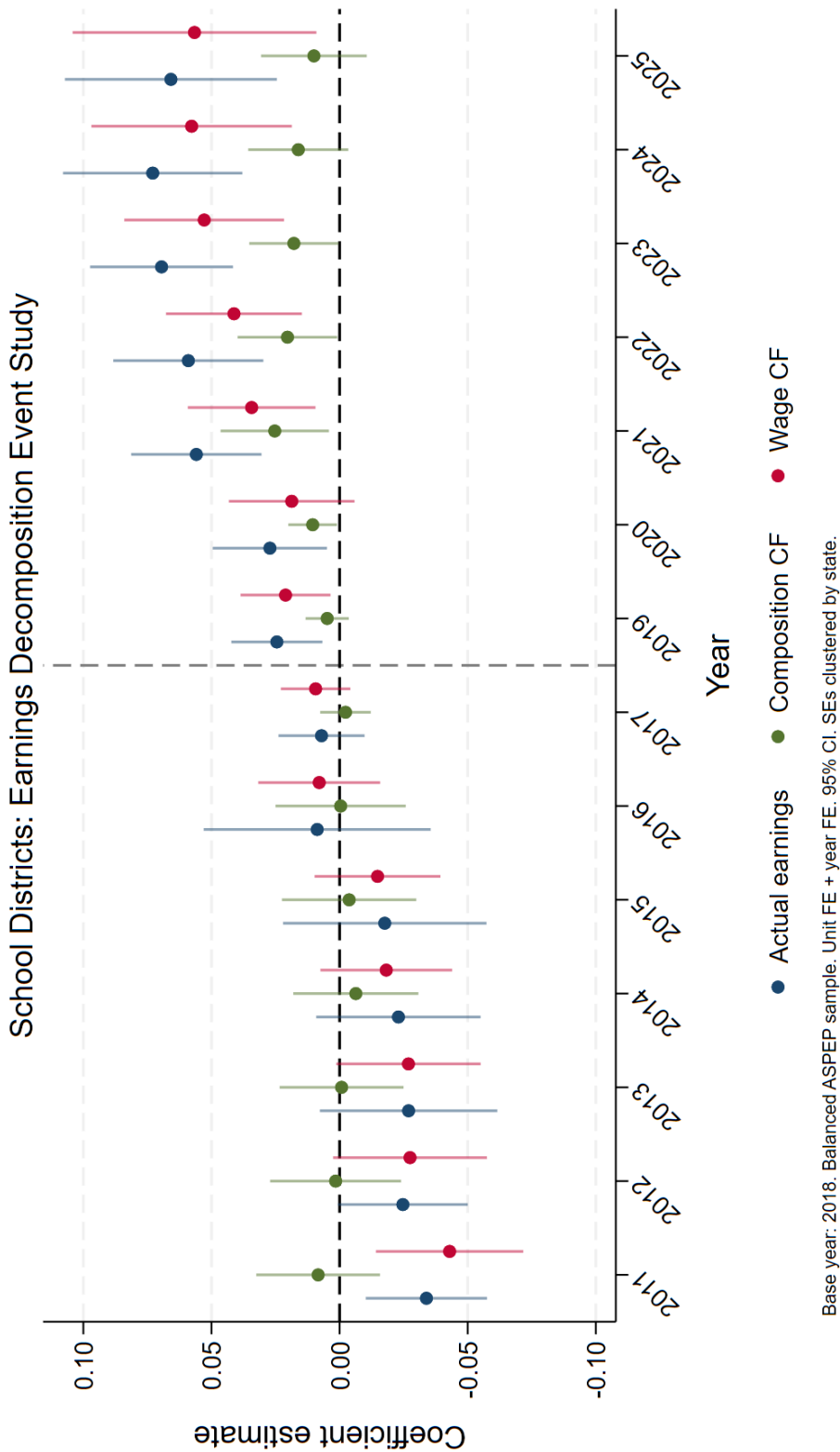
Figure A.1: Event study plot of the effects of *Janus* on PT employment and FT share in the private-sector using QCEW data

Event Study: Private-Sector Employment Composition (QCEW placebo — *Janus* v. *AFSCME*)



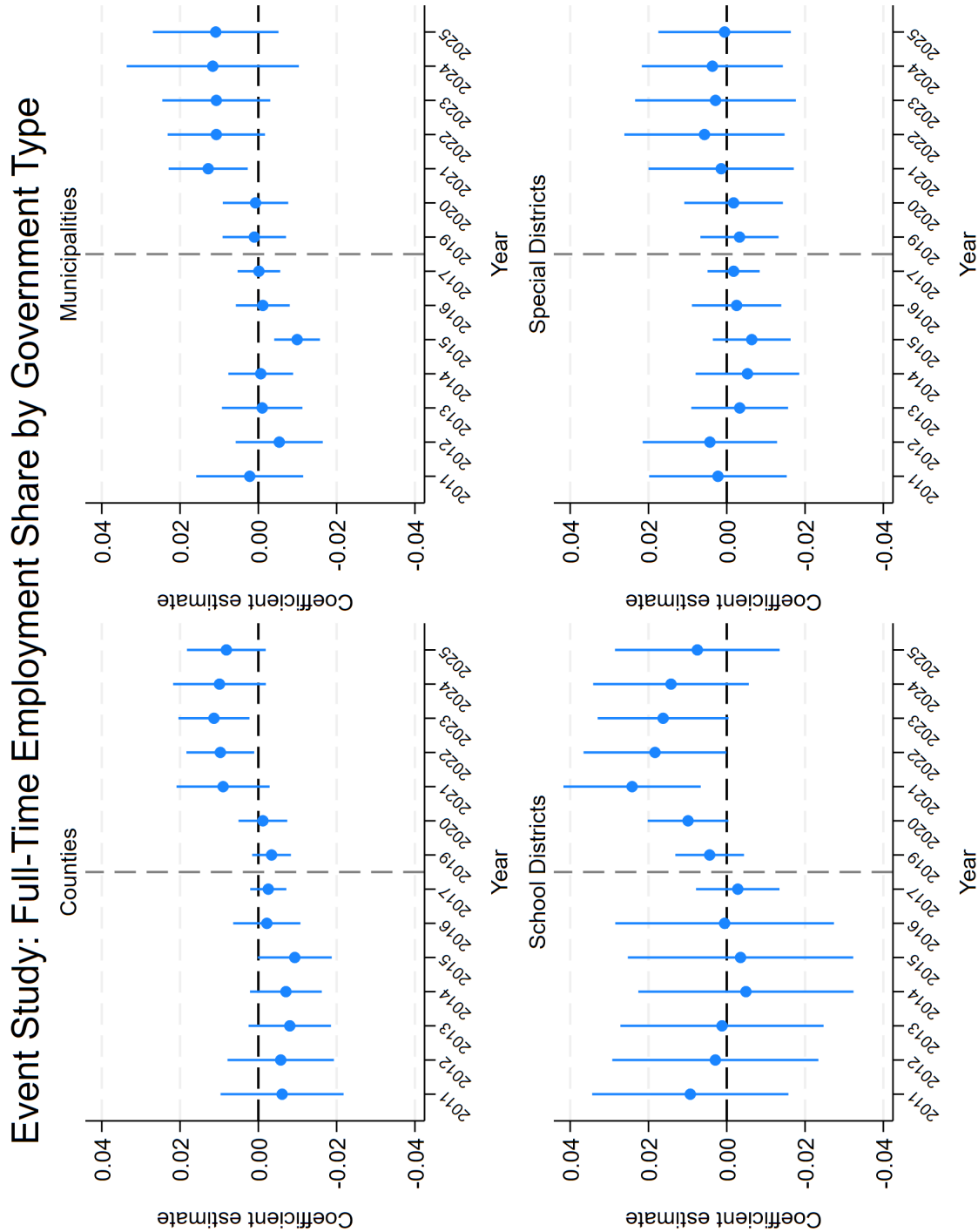
Notes: Each panel plots event-study coefficients from a regression of the outcome on interactions of the treatment indicator with year dummies, relative to the base year 2018. The sample consists of 47 states (excluding Colorado, Missouri, and Wisconsin). The regressions include state fixed effects and year fixed effects. The treatment indicator equals one for states that had a mandatory agency-fee law at the time of the *Janus* ruling. The outcome is measured using BLS Quarterly Census of Employment and Wages (QCEW) state-level annual averages for the private sector. The dotted vertical line marks June 2018 (*Janus* decision). Horizontal dashed line at zero. Regressions are unweighted. The figure displays point estimates and 95 percent confidence intervals; standard errors are clustered by state. The 2020 coefficient for log private part-time employment reflects differential pandemic-era disruption — treated states (e.g., California, Illinois, and New York) imposed significantly stricter lockdown policies that disproportionately contracted employment in the private-sector — rather than a *Janus* effect.

Figure A.2: Event-study Decomposition of earnings increase – compositional counterfactual (holding wages steady) and wage counterfactual (holding composition steady) for school districts



Notes: The figure plots event-study coefficients for three outcomes for Independent School Districts only. “Actual earnings” is log total per-worker annual earnings. “Composition effect” is the log earnings counterfactual holding wages fixed at their 2017 values while allowing the full-time/part-time mix to evolve; a non-zero coefficient indicates a composition effect. “Wages effect” is the log earnings counterfactual holding the 2017 full-time/part-time mix fixed while allowing wages to change; a non-zero coefficient indicates a wage effect. The sample is the CoG-ASPEP annual balanced panel (2011–2025) restricted to Independent School Districts, observed in all 15 years. Base year is 2018; regressions include government-unit fixed effects and year fixed effects. The treatment indicator equals one for school districts in agency-fee states at the time of *Janus*. Vertical dashed line marks June 2018. Regressions are unweighted. Shaded bands are 95% confidence intervals; standard errors clustered by state.

Figure A.3: Event Study: Full-Time Employment Share by Government Type



Base year: 2018. Balanced ASPEP sample. Unit FE + year FE. 95% CI. SEs clustered by state.

Notes: The figure plots event-study coefficients for the share of full-time workers in total employment estimated separately for each government type. Coefficients are estimated on the interactions of the treatment indicator with year dummies, with 2018 as the omitted base year. Pre-period coefficients (2011–2017) serve as a test of parallel pre-trends; post-period coefficients (2019–2025) measure the differential change in governments in treated states relative to governments in control states following the *Janus* ruling. Data are from the CoG-ASPEP annual balanced panel (2011–2025); townships are excluded. The underlying regressions include government-unit fixed effects and year fixed effects and are unweighted. The figure displays point estimates and 95 percent confidence intervals; standard errors are clustered by state.

Table A.1: Summary Statistics for the Estimation Sample for the CoG three-wave survey (2012, 2017, and 2022)

	Mean	Median	SD	N
Panel A: Overall sample				
Full-time employment	378.7	40.0	4617.7	34898
Part-time employment	121.8	16.0	1639.1	34898
Total employment	500.5	60.0	6080.2	34898
Full-time annual earnings	50703	47627	18032	34898
Part-time annual earnings	12194	10740	8045	34898
Total earnings	36723	35391	16631	34898
Share of full-time employment	0.64	0.69	0.24	34898
Panel B: Counties				
Full-time employment	855.2	198.5	3226.6	2730
Part-time employment	172.9	38.0	593.3	2730
Total employment	1028.2	242.0	3680.8	2730
Full-time annual earnings	44714	43099	11313	2730
Part-time annual earnings	13850	12510	6759	2730
Total earnings	38877	37354	10952	2730
Share of full-time employment	0.81	0.83	0.12	2730
Panel C: Municipalities				
Full-time employment	174.5	16.0	3932.0	12693
Part-time employment	38.6	12.0	343.3	12693
Total employment	213.1	30.0	4264.8	12693
Full-time annual earnings	49698	46792	18951	12693
Part-time annual earnings	9987	8862	6752	12693
Total earnings	33011	30925	17894	12693
Share of full-time employment	0.57	0.60	0.26	12693
Panel D: School Districts				
Full-time employment	404.9	149.0	1260.6	11249
Part-time employment	143.7	44.0	448.1	11249
Total employment	548.6	203.0	1601.1	11249
Full-time annual earnings	52107	48888	15720	11249
Part-time annual earnings	13660	12237	7656	11249
Total earnings	40722	38569	12001	11249
Share of full-time employment	0.72	0.75	0.17	11249
Panel E: Special Districts				
Full-time employment	58.2	7.0	472.4	8179
Part-time employment	20.7	4.0	141.9	8179
Total employment	78.9	13.0	598.9	8179
Full-time annual earnings	52265	48756	20678	8179
Part-time annual earnings	13032	11082	9864	8179
Total earnings	36197	32418	19972	8179
Share of full-time employment	0.59	0.62	0.26	8179

Notes: The table reports means, standard deviations, and sample sizes for the Census of Governments three-wave estimation sample (survey years 2012, 2017, and 2022), separately by treatment group. Treated units are local governments in states that had a mandatory agency-fee law at the time of the *Janus* ruling; control units are in states without such a law. Colorado, Missouri, and Wisconsin are excluded because their laws changed during the sample period. Townships are excluded. The regression sample is smaller than the full CoG universe because of singleton dropping and the requirement that units appear in all three survey waves. Employment figures are in persons; earnings figures are annual pay in dollars. FT share is the ratio of full-time to total employees.

Table A.2: Summary Statistics for the Estimation Sample for the CoG – ASPEP annual balanced panel (2011 – 2025)

	Mean	Median	SD	N
Panel A: Overall sample				
Full-time employment	2607.9	468.5	14028.0	3670
Part-time employment	775.0	93.0	4997.3	3670
Total employment	3382.9	602.5	18478.9	3670
Full-time annual earnings	57956	54621	16637	3670
Part-time annual earnings	15708	13633	9343	3670
Total earnings	48312	46013	15173	3670
Share of full-time employment	0.78	0.82	0.16	3670
Panel B: Counties				
Full-time employment	2421.7	894.0	5821.1	757
Part-time employment	455.3	118.0	1060.1	757
Total employment	2877.0	993.0	6608.1	757
Full-time annual earnings	52664	50976	12132	757
Part-time annual earnings	17209	15555	8164	757
Total earnings	46974	45459	11262	757
Share of full-time employment	0.84	0.85	0.10	757
Panel C: Municipalities				
Full-time employment	1276.2	292.0	12001.1	1350
Part-time employment	197.9	64.0	1035.1	1350
Total employment	1474.1	369.0	13008.8	1350
Full-time annual earnings	61609	59342	16186	1350
Part-time annual earnings	12309	11089	6406	1350
Total earnings	50456	49176	14751	1350
Share of full-time employment	0.78	0.82	0.15	1350
Panel D: School Districts				
Full-time employment	2666.0	1555.0	4063.8	769
Part-time employment	845.5	495.0	1429.1	769
Total employment	3511.5	2264.0	5070.7	769
Full-time annual earnings	55488	51327	15919	769
Part-time annual earnings	15746	14672	7190	769
Total earnings	43365	41125	10147	769
Share of full-time employment	0.73	0.77	0.17	769
Panel E: Special Districts				
Full-time employment	440.4	134.0	1456.9	747
Part-time employment	109.2	22.0	441.7	747
Total employment	549.6	184.0	1851.8	747
Full-time annual earnings	58986	54819	20276	747
Part-time annual earnings	20313	16808	13740	747
Total earnings	50893	47695	21416	747
Share of full-time employment	0.79	0.85	0.20	747

Notes: The table reports means, standard deviations, and sample sizes for the CoG–ASPEP annual balanced panel (2011–2025), separately by treatment group. The balanced panel requires each unit to be observed in all 15 years (2011–2025); the regression sample is smaller due to singleton dropping. Treated units are local governments in states that had a mandatory agency-fee law at the time of the *Janus* ruling; control units are in states without such a law. Colorado, Missouri, and Wisconsin are excluded. Townships are excluded. Employment figures are in persons; earnings figures are annual pay in dollars per worker. FT share is the ratio of full-time to total employees.

Table A.3: Robustness Check: Estimates of the impact of *Janus* on employment and earnings using alternative specifications

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Using TWFE and the three-wave Census of Governments (CoG) for 2012, 2017, and 2022							
Dependent variable	FT Empl.	PT Empl.	Total Empl.	FT Earnings	PT Earnings	Total Earnings	Full-time share
Coefficient on interaction term, Treated X Post	-0.00149 (0.0165)	-0.0754*** (0.0227)	-0.0282* (0.0155)	0.000831 (0.0100)	-0.00414 (0.0111)	0.0174* (0.00902)	0.0140*** (0.00385)
Number of observations	102241	102241	102241	102241	102241	102241	102241
Number of clusters	47	47	47	47	47	47	47
Panel B: Using TWFE and the balanced panel constructed from the CoG and ASPEP spanning 2011 – 2025							
Coefficient on interaction term, Treated X Post	-0.000362 (0.0172)	-0.0617* (0.0312)	-0.0183 (0.0149)	-0.00136 (0.00910)	0.0226 (0.0201)	0.0100 (0.00979)	0.0115*** (0.00425)
Number of observations	55029	55029	55029	55029	55029	55029	55029
Number of clusters	47	47	47	47	47	47	47
Panel C: Using IHS transformation and the three-wave Census of Governments (CoG) for 2012, 2017, and 2022							
Coefficient on interaction term, Treated X Post	0.0085 (0.0110)	-0.0858*** (0.0172)	-0.0267** (0.0090)	-0.0047 (0.0075)	0.0040 (0.0100)	0.0183* (0.0077)	0.0153*** (0.0033)
Number of observations	100597	100597	100597	100597	100597	100597	100597
Number of clusters	47	47	47	47	47	47	47
Panel D: Using IHS transformation and the balanced panel constructed from the CoG and ASPEP spanning 2011 – 2025							
Coefficient on interaction term, Treated X Post	0.0045 (0.0117)	-0.0625* (0.0259)	-0.0138 (0.0101)	-0.0044 (0.0076)	0.0211 (0.0180)	0.0081 (0.0083)	0.0095** (0.0033)
Number of observations	55029	55029	55029	55029	55029	55029	55029
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes:The dependent variables are the log or inverse hyperbolic sine (IHS) transformation of full-time employment (FT Empl.), part-time employment (PT Empl.), total employment (Tot Empl.), annualized full-time payroll per worker (FT Earnings), part-time payroll per worker (PT Earnings), total payroll per worker (Tot Earnings), and the share of full-time workers in total employment (FT Share). The treatment indicator equals one if the unit is located in a state that had a mandatory agency-fee law in place at the time of the *Janus* ruling (June 2018) and zero otherwise; Colorado, Missouri, and Wisconsin are excluded because their laws changed during the sample period. Panels A and B use the natural logarithm of each outcome and a two-way fixed effects (TWFE) specification with government-unit and year fixed effects only, without state-specific pre-period trends. Panels C and D replace the log with the IHS transformation $-IHS(x) = \ln(x + \sqrt{1+x^2})$ — which equals zero when $x = 0$ and approximates $\ln(2x)$ for large x , retaining zero-employment and zero-earnings observations that are excluded by the log; these panels add state-specific pre-period linear trends to the TWFE specification. Panels A and C use the Census of Governments three-wave panel (survey years 2012, 2017, and 2022); Treated X Post equals one in 2022 for treated units. Panels B and D use the CoG-ASPEP annual balanced panel (2011–2025); Treated X Post equals one in 2019 and later for treated units. All panels exclude townships. The common sample in each panel is the set of observations with non-missing values across all seven outcomes, determined by the part-time employment specification under TWFE with state-specific pre-period trends. All regressions are unweighted.

Table A.4: Weighted estimates of the impact of *Janus* on employment and earnings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Using the three-wave Census of Governments (CoG) for 2012, 2017, and 2022							
Dependent variable	FT Empl.	PT Empl.	Total Empl.	FT Earnings	PT Earnings	Total Earnings	Full-time share
Coefficient on interaction term, Treated X Post	0.00329 (0.0123)	-0.0138 (0.0188)	-0.0143 (0.00912)	-0.00376 (0.00672)	0.0126 (0.0248)	0.00858 (0.00840)	0.00935*** (0.00444)
Number of observations	102179	102179	102179	102179	102179	102179	102179
Number of clusters	47	47	47	47	47	47	47
Panel B: Using the balanced panel constructed from the CoG and ASPEP spanning 2011 – 2025							
Coefficient on interaction term, Treated X Post	0.00834 (0.0150)	0.0226 (0.0224)	-0.00103 (0.0109)	0.00437 (0.00719)	-0.0207 (0.0301)	0.00909 (0.00967)	0.00505 (0.00486)
Number of observations	55029	55029	55029	55029	55029	55029	55029
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes:The dependent variables are the natural logarithm of full-time employment (FT Empl.), part-time employment (PT Empl.), and total employment (Total Empl.); the natural logarithm of annualized full-time payroll per worker (FT Earnings), part-time payroll per worker (PT Earnings), and total payroll per worker (Tot Earnings); and the share of full-time workers in total employment (FT Share). The treatment indicator equals one if the unit is located in a state that had a mandatory agency-fee law in place at the time of the *Janus* ruling (June 2018) and zero otherwise; Colorado, Missouri, and Wisconsin are excluded because their laws changed during the sample period. Panel A uses the Census of Governments three-wave panel (survey years 2012, 2017, and 2022); Treated X Post equal to one in 2022 for treated units. Panel B uses the CoG-ASPEP annual balanced panel (2011–2025); Treated X Post equal to one in 2019 and later for treated units. Both panels exclude townships. The regression sample in each panel is the common sample defined as observations with non-missing values across all seven outcomes, determined by the full-time employment share specification. All regressions include government-unit fixed effects, year fixed effects, and state-specific pre-period linear trends. Regressions are weighted.

Table A.5: Difference-in-Differences estimates of the impact of *Janus* on employment and earnings using border-state matching and covariate matching approaches on the CoG-ASPEP balanced dataset

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Using a border-state matching approach							
Dependent variable	FT empl.	PT empl.	Total empl.	FT earnings	PT earnings	Total earnings	Full-time share
Coefficient on interaction term, Treated X Post	-0.0120 (0.0210)	-0.0839* (0.0400)	-0.0321 (0.0207)	0.00771 (0.0116)	0.0342 (0.0269)	0.0226** (0.0106)	0.0111* (0.00618)
Wild-cluster p-value	0.608	0.052	0.163	0.520	0.248	0.033	0.085
Number of observations	6336	6336	6336	6336	6336	6336	6336
Number of clusters	23	23	23	23	23	23	23
Panel B: Using log of long-term debt per employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	-0.0120 (0.0147)	-0.0791* (0.0400)	-0.0245 (0.0152)	-0.0160* (0.00824)	0.00375 (0.0266)	-0.00807 (0.00818)	0.00975** (0.00481)
Number of observations	16304	16304	16304	16304	16304	16304	16304
Number of clusters	42	42	42	42	42	42	42
Panel C: Using log of cash reserves and deposits per employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	-0.00155 (0.0131)	-0.0419 (0.0314)	-0.0184 (0.0145)	-0.0146* (0.00754)	0.000755 (0.0219)	-0.00310 (0.00848)	0.00975** (0.00361)
Number of observations	20750	20750	20750	20750	20750	20750	20750
Number of clusters	42	42	42	42	42	42	42
Panel D: Using log of long-term debt and cash reserves/ employee as of 2017 for matching units in treated and control states							
Coefficient on interaction term, Treated X Post	-0.00144 (0.0145)	-0.0647* (0.0352)	-0.0183 (0.0160)	-0.0197** (0.00870)	-0.00813 (0.0249)	-0.00902 (0.00802)	0.0110** (0.00427)
Number of observations	15583	15583	15583	15583	15583	15583	15583
Number of clusters	42	42	42	42	42	42	42

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: All panels use the CoG-ASPEP annual balanced panel (2011–2025), excluding townships. Treated X Post equal to one in 2019 and later for treated units. The dependent variables are the same seven outcomes as in Table 1. Panel A (state-border-pair DiD) restricts to government units in counties that straddle a treated/control state border and includes unit fixed effects, year fixed effects, and state-border-pair segment fixed effects; the comparison group is border-county units on the control side of each state pair. Significance stars are based on wild cluster bootstrap (WCB) p-values using Webb (2023) weights, 999 replications, seed 12345; asymptotic standard errors clustered by state are shown in parentheses for reference. Panels B–D (matched DiD) use 1-nearest-neighbor Mahalanobis matching exact on government type; matching covariates vary by sub-panel: long-run solvency (Panel B), cash solvency (Panel C), or both (Panel D). Matched DiD estimates are shown for the preferred specification with two-way fixed effects (unit and year fixed effects) and state-specific pre-period linear trends, matching the baseline approach in Table 1. The regression sample in each panel is the set of observations with non-missing values for all seven outcomes, determined by the full-time-share specification. All regressions are unweighted.

Table A.6: Event study estimates of the impact of *Janus* on FT employment, PT employment, and FT share using the annual balanced panel

	(1)	(2)	(3)
Dependent variable	FT employment	PT employment	Full-time share
2011	0.0156 (0.0170)	0.00729 (0.0335)	0.00136 (0.00481)
2012	0.00472 (0.0163)	0.00306 (0.0356)	-0.00177 (0.00510)
2013	-0.00417 (0.0157)	-0.00961 (0.0346)	-0.00288 (0.00469)
2014	-0.00842 (0.0132)	-0.00606 (0.0301)	-0.00417 (0.00440)
2015	-0.00887 (0.0112)	0.0249 (0.0312)	-0.00775* (0.00419)
2016	-0.00830 (0.00959)	-0.0108 (0.0262)	-0.00163 (0.00378)
2017	-0.00164 (0.00418)	0.0138 (0.0138)	-0.00163 (0.00211)
2019	-0.00394 (0.00432)	-0.0106 (0.0138)	-0.0000942 (0.00189)
2020	-0.00716 (0.00562)	-0.0290 (0.0197)	0.00194 (0.00224)
2021	-0.0114 (0.00790)	-0.0960*** (0.0341)	0.0120** (0.00473)
2022	-0.00250 (0.00972)	-0.0760** (0.0286)	0.0112** (0.00464)
2023	-0.00212 (0.0112)	-0.0736** (0.0314)	0.0105** (0.00482)
2024	0.00798 (0.0134)	-0.0747* (0.0382)	0.0107* (0.00585)
2025	0.00553 (0.0133)	-0.0524 (0.0384)	0.00791 (0.00485)
Constant	6.071*** (0.00149)	4.543*** (0.00736)	0.800*** (0.00109)
Number of observations	58414	55047	58431
Number of clusters	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: The table reports year-by-year event-study coefficients corresponding to Figure 1, estimated on the CoG-ASPEP annual balanced panel (2011–2025). Each coefficient is the interaction of the treatment indicator with a year dummy, relative to the base year 2018. The outcomes are log full-time employment, log part-time employment, and the full-time employment share. The treatment indicator equals one for local governments in states that had a mandatory agency-fee law at the time of the *Janus* ruling. All regressions include government-unit fixed effects and year fixed effects. Regressions are unweighted.

Table A.7: Difference-in-Differences estimates of the impact of *Janus* on employment and earnings – Split by type of government using the annual balanced panel

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Counties							
Dependent variable	FT empl.	PT empl.	Total empl.	FT earnings	PT earnings	Total earnings	Full-time share
Coefficient on interaction term, Treated X Post	-0.0192 (0.0151)	-0.0854** (0.0332)	-0.0274* (0.0156)	-0.00525 (0.00895)	0.0277 (0.0274)	0.00250 (0.00929)	0.00788** (0.00386)
Number of observations	11304	11304	11304	11304	11304	11304	11304
Number of clusters	45	45	45	45	45	45	45
Panel B: Municipalities							
Coefficient on interaction term, Treated X Post	-0.00317 (0.0152)	-0.0899* (0.0503)	-0.0167 (0.0173)	-0.0219* (0.0120)	0.0195 (0.0256)	-0.0116 (0.00900)	0.0106* (0.00617)
Number of observations	20309	20309	20309	20309	20309	20309	20309
Number of clusters	47	47	47	47	47	47	47
Panel C: School Districts							
Coefficient on interaction term, Treated X Post	0.0171 (0.0145)	0.00731 (0.0594)	-0.0105 (0.0119)	0.0308 (0.0195)	0.00493 (0.0352)	0.0484** (0.0201)	0.0142 (0.00864)
Number of observations	11462	11462	11462	11462	11462	11462	11462
Number of clusters	41	41	41	41	41	41	41
Panel D: Special Districts							
Coefficient on interaction term, Treated X Post	0.0210 (0.0177)	-0.0486 (0.0430)	0.00108 (0.0167)	-0.00687 (0.0112)	0.0323 (0.0348)	0.00462 (0.0131)	0.00900 (0.00671)
Number of observations	11249	11249	11249	11249	11249	11249	11249
Number of clusters	47	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: Each panel presents results from a specification with two-way fixed effects and state-specific pre-period linear trends for a single government type, estimated separately within that type, using the CoG-ASPEP annual balanced panel (2011–2025). Treated X Post equal to one in 2019 and later for treated units. Government types are County (A), Municipality (B), Independent School District (C), and Special District (D). Townships and state-government units are excluded. Within each type, the regression sample is the set of observations with non-missing values for all seven outcomes, determined by the full-time-share specification. All regressions include government-unit fixed effects, year fixed effects, and state-specific pre-period linear trends. Regressions are unweighted.

Table A.8: The impact of *Janus* on earnings and hourly wages using the Current Population Survey

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dep. variable – Earnings (as-is in Cols. (1) – (3)); winsorized (in cols. (4) – (6))						
Coefficient on interaction term, Treated X Post	0.0160 (0.0180)	0.00735 (0.0183)	0.0120 (0.0169)	0.0115 (0.0149)	0.00380 (0.0143)	0.00775 (0.0134)
Dummy variable for female		-0.361*** (0.0113)	-0.286*** (0.00590)		-0.334*** (0.00882)	-0.271*** (0.00432)
Dummy variable for High School diploma		0.366*** (0.0291)	0.255*** (0.0216)		0.309*** (0.0263)	0.215*** (0.0206)
Dummy variable for some college		0.500*** (0.0360)	0.398*** (0.0248)		0.446*** (0.0330)	0.359*** (0.0241)
Dummy variable for a Bachelor's degree		0.781*** (0.0441)	0.623*** (0.0303)		0.700*** (0.0401)	0.566*** (0.0295)
Dummy variable for a graduate degree		1.034*** (0.0443)	0.864*** (0.0290)		0.941*** (0.0394)	0.795*** (0.0277)
Age		0.0994*** (0.00362)	0.0643*** (0.00257)		0.0825*** (0.00282)	0.0527*** (0.00194)
Age squared		-0.00101*** (0.0000363)	-0.000629*** (0.0000260)		-0.000833*** (0.0000284)	-0.000509*** (0.0000199)
Dummy variable for full-time worker			0.889*** (0.0143)			0.756*** (0.0126)
Number of observations	98787	98787	98787	98787	98787	98787
Number of clusters	47	47	47	47	47	47
Panel B: Dep. variable – Wages (as-is in Cols. (1) – (3)); winsorized (in cols. (4) – (6))						
Coefficient on interaction term, Treated X Post	0.0118 (0.0152)	0.00596 (0.0152)	0.00653 (0.0151)	0.0119 (0.0127)	0.00649 (0.0122)	0.00704 (0.0121)
Dummy variable for female		-0.254*** (0.00659)	-0.245*** (0.00563)		-0.237*** (0.00527)	-0.228*** (0.00427)
Dummy variable for High School diploma		0.231*** (0.0236)	0.217*** (0.0230)		0.199*** (0.0220)	0.186*** (0.0212)
Dummy variable for some college		0.378*** (0.0272)	0.365*** (0.0259)		0.336*** (0.0260)	0.324*** (0.0246)
Dummy variable for a Bachelors' degree		0.597*** (0.0321)	0.577*** (0.0303)		0.543*** (0.0305)	0.524*** (0.0285)
Dummy variable for a graduate degree		0.820*** (0.0324)	0.799*** (0.0302)		0.753*** (0.0305)	0.733*** (0.0282)
Age		0.0499*** (0.00241)	0.0456*** (0.00204)		0.0437*** (0.00217)	0.0396*** (0.00179)
Age squared		-0.000474*** (0.0000248)	-0.000428*** (0.0000211)		-0.000414*** (0.0000229)	-0.000369*** (0.0000190)
Dummy variable for full-time worker			0.109*** (0.0144)			0.105*** (0.0129)
Number of observations	98789	98789	98789	98789	98789	98789
Number of clusters	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: The data are from the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC). The sample is restricted to public-sector workers (state or local government employees) who report positive earnings. The treatment indicator equals one for workers in states that had a mandatory agency-fee law at the time of the *Janus* ruling, interacted with a post-2018 indicator. Panel A uses log annual earnings; Panel B uses log hourly wages. For each outcome, three covariate specifications are shown: parsimonious (no controls), baseline (female, education dummies, age, age squared), and full (baseline plus full-time indicator). Columns (4)–(6) use 5th/95th-percentile winsorized versions of the wage variables. All regressions include state fixed effects, year fixed effects, and state-specific pre-period linear trends. The regression sample within each panel is determined by the most demanding (full-covariate) specification.

Table A.9: The impact of *Janus* on provision of benefits using the Current Population Survey

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Dep. var. – Employer health insurance coverage (LPM in Cols. (1) – (3); probit in cols. (4) – (6))						
Coefficient on interaction term, Treated X Post	0.00735 (0.0128)	0.00616 (0.0129)	0.00966 (0.0124)	0.00456 (0.0430)	0.00583 (0.0446)	0.0393 (0.0453)
Dummy variable for female		-0.0720*** (0.00411)	-0.0453*** (0.00319)		-0.266*** (0.0144)	-0.178*** (0.0125)
Dummy variable for High School diploma		0.142*** (0.0157)	0.0969*** (0.0141)		0.405*** (0.0479)	0.301*** (0.0490)
Dummy variable for some college		0.158*** (0.0166)	0.116*** (0.0142)		0.468*** (0.0508)	0.379*** (0.0494)
Dummy variable for a Bachelor's degree		0.207*** (0.0197)	0.144*** (0.0164)		0.632*** (0.0602)	0.477*** (0.0573)
Dummy variable for a graduate degree		0.239*** (0.0202)	0.172*** (0.0166)		0.768*** (0.0628)	0.599*** (0.0584)
Age		0.0294*** (0.000979)	0.0158*** (0.000930)		0.0961*** (0.00297)	0.0559*** (0.00303)
Age squared		-0.000301*** (0.0000108)	-0.000153*** (0.0000101)		-0.000986*** (0.0000331)	-0.000546*** (0.0000335)
Dummy variable for full-time worker			0.304*** (0.00629)			0.917*** (0.0177)
Number of observations	101521	101521	101521	101521	101521	101521
Number of clusters	47	47	47	47	47	47
Panel B: Dep. var. – Pension offered (LPM in Cols. (1) – (3); probit in cols. (4) – (6))						
Coefficient on interaction term, Treated X Post	-0.00174 (0.0125)	-0.00276 (0.0115)	-0.000925 (0.0118)	0.0423 (0.0355)	0.0388 (0.0351)	-0.00632 (0.0354)
Dummy variable for female		-0.0367*** (0.00356)	-0.0226*** (0.00312)		-0.115*** (0.0112)	-0.0732*** (0.00984)
Dummy variable for High School diploma		0.158*** (0.0174)	0.134*** (0.0170)		0.411*** (0.0452)	0.353*** (0.0443)
Dummy variable for some college		0.207*** (0.0226)	0.185*** (0.0217)		0.556*** (0.0594)	0.505*** (0.0575)
Dummy variable for a Bachelors' degree		0.263*** (0.0236)	0.230*** (0.0225)		0.725*** (0.0629)	0.643*** (0.0603)
Dummy variable for a graduate degree		0.294*** (0.0231)	0.259*** (0.0220)		0.839*** (0.0613)	0.749*** (0.0585)
Age		0.0245*** (0.000681)	0.0173*** (0.000806)		0.0707*** (0.00224)	0.0502*** (0.00249)
Age squared		-0.000243*** (0.00000732)	-0.000165*** (0.00000865)		-0.000701*** (0.0000237)	-0.000478*** (0.0000266)
Dummy variable for full-time worker			0.160*** (0.00539)			0.460*** (0.0152)
Number of observations	101521	101521	101521	101521	101521	101521
Number of clusters	47	47	47	47	47	47

Standard errors clustered by state in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Notes: The data are from the Current Population Survey Annual Social and Economic Supplement (CPS-ASEC). The sample is the full CPS-ASEC sample restricted to public-sector workers (state or local government employees), including those who do not report earnings. The treatment indicator equals one for workers in states that had a mandatory agency-fee law at the time of the *Janus* ruling, interacted with a post-2018 indicator. Panel A uses an indicator for employer-sponsored health insurance coverage in the prior year; Panel B uses an indicator for whether the employer offered a pension plan. For each outcome, three LPM specifications (parsimonious, baseline controls, full controls) and three probit specifications (same covariate sets) are shown. The probit specification in column (6) also includes state-specific pre-period linear trends. The regression sample is determined by the full-covariate LPM specification, estimated jointly across both benefit outcomes.

Table A.10: Public-Sector Employment and Earnings: Evidence from the BLS Quarterly Census of Employment and Wages (QCEW)

	(1)	(2)	(3)	(4)
Panel A: Log public-sector employment				
Dependent variable	Employment at the state level		Employment at the local level	
Coefficient on interaction term, Treated X Post	-0.0259 (0.0220)	-0.0139 (0.0141)	-0.0106 (0.0129)	-0.0133 (0.0077)
Number of observations	658	658	658	658
Number of clusters	47	47	47	47
Panel B: Log average annual pay				
	Earnings of state workers		Earnings of local workers	
Coefficient on interaction term, Treated X Post	0.0272 (0.0201)	0.0226 (0.0177)	0.0019 (0.0117)	-0.0047 (0.0103)
Number of observations	658	658	658	658
Number of clusters	47	47	47	47

Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Notes: Each cell reports the coefficient on Treated $\times$ Post from a difference-in-differences regression estimated on a state-level panel, 2011–2024. The dependent variable in Panel A is log annual average employment; in Panel B, log average annual pay (average weekly wage $\times$ 52). Columns (1) and (3) report estimates from a two-way fixed effects (TWFE) specification with state and year fixed effects. Columns (2) and (4) add state-specific linear pre-period trends (the baseline specification). The unit of observation is a state-year. Post = 1 for years 2019 and later. Treatment is defined identically to Table 1.

The QCEW data are drawn from the BLS annual singlefile releases for 2011–2024. For each year, I retain the statewide ownership-total record (aggregation level 51, industry code 10) separately for state government (ownership code 2) and local government (ownership code 3). This aggregation level is the only one that is unsuppressed for all states; sub-ownership aggregation levels (52 and 53) are suppressed for several large treated states. Observations with zero reported employment or zero reported wages are dropped. The treatment indicator and all remaining exclusions are identical to those in Table 1.